

COMP.2560: System Programming

LAB #3: More Debugging

LAB 3: The `man` command

The complete documentation for any Unix/Linux system, known as `man` pages (manual pages), is available to each user via the “`man`” command. Whenever you have doubts about how to use a C function or any (shell) commands, besides googling it online, you can get detailed documentation for that function or command using the `man` command.

Visit this link: <https://www.geeksforgeeks.org/man-command-in-linux-with-examples/>, which contains a very brief article to get you started.

Try it yourself on any function we have studied, such as `open()`, `close()`, `read()`, `write()`, `fopen()`, `fclose()`, etc.

You need to know how to effectively use the “`man`” command yourself whenever you have a question about C functions or Linux commands.

LAB 3: Overview

The main purpose of this lab is to:

- Part I: If needed, customize `gdb` a little bit so that it works smoothly (**no submission required for this part**).
- Part II: Debug the program “`filesize1v.c`”. **Submission is required**.
- Part III: This is a programming question and **submission is required**.
See the description in Part III below for details.

These three parts of the lab are described below.

LAB 3 – Part I: Configuring the Debugger

- Watch the video (“L.3. COMP.2560 - Debugging (cont.).mp4”), to decide if you need to do step 2 below.
- According to the instructions in the video, copy the attached file (“`.gdbinit`”) to your home directory on CS server. Then try to use `gdb` to see if it solves the problem demonstrated in the video.

No submission is required for Part I.

LAB 3 – Part II: GDB Debugging your program

Recall that in class, we discussed the code `filesize1.c` and explained why it does not work. In this part, a change has been made to one line in `filesize1.c`, that line is

```
char c;
```

which has been changed to

```
unsigned char c;
```

hoping that the program will now work properly.

Review the attached code `filesize1v.c`, which is the same as `filesize1.c` except for this single line of change as described above.

Compile and run `filesize1v.c` with `example.txt` file provided and you will find it is still not working as expected. Use `gdb` to find out what the problem is with this code.

Write a small text file to briefly explain what the problem is and submit it.

LAB 3 – Part III: Rewrite a program

1. Review the code in file `p16.c` to fully understand how it works and what it does.
2. Rewrite `p16.c` using system I/O functions (`open()`, `close()`, `read()`, and `write()`).
Ask your lab instructors for help if any questions.

LAB 3 – Submission

Submit the items elements described above.

To formally submit your work for this Lab:

- Submit a text file, briefly explaining what your findings are for Part II.
- Submit the re-written code for `p16.c`.

When submitting, please indicate in the submission textbox the Lab section you belong to.

Evaluation:

1. It is our intention to evaluate your work being done in-person, while you are in the lab.
2. It is strongly suggested that you attend in-person to familiarize yourselves with the Lab infrastructure, meet with the GATAs and solve problems.
3. Each session is only 80 minutes long, so it is strongly suggested that you arrive early, are focused and do not disrupt the process.
4. It is desired that you complete your work while in the lab and get graded after live review from the GATAs.
5. Should you need more time, you can take your work home and present it to the GATAs on the very next lab session (the week following).
6. Even if you decide to work from home and submit your work remotely, you still need to have your work peer reviewed, in-person.

NOTE: From this Lab and on, **NO** video recording will be required, unless specifically stated differently. You need to submit only what is required by the lab. You **DO NEED** to have your work peer reviewed in the lab, **IN-PERSON**.